Divergence of a Vector Field

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9. Vector Fields

9.1 A vector field $\vec{v}(x,y,z)$ on $\mathbb{R}^3$ is a vector where each component is a scalar field, i.e.,

$$\vec{v}(x,y,z) = \begin{bmatrix} f(x,y,z) \\ g(x,y,z) \\ h(x,y,z) \end{bmatrix}.$$

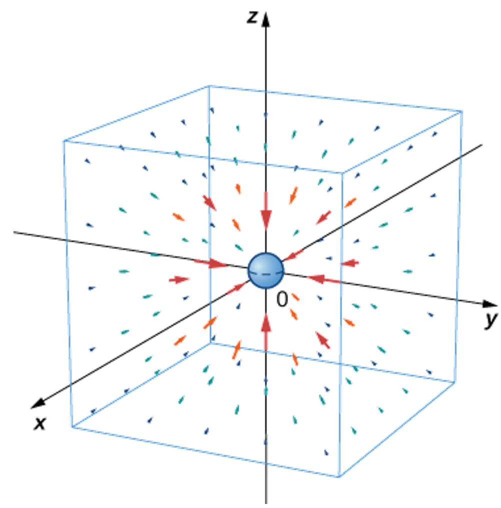
In other words,

$$\vec{v} = f(x,y,z)\,\hat{\imath} + g(x,y,z)\,\hat{\jmath} + h(x,y,z)\,\hat{k}$$

9.2 Examples of Fields that you may have seen:

(1) Gravitational

$$\vec{F} = \frac{-GM}{(x^2+y^2+z^2)^{3/2}} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

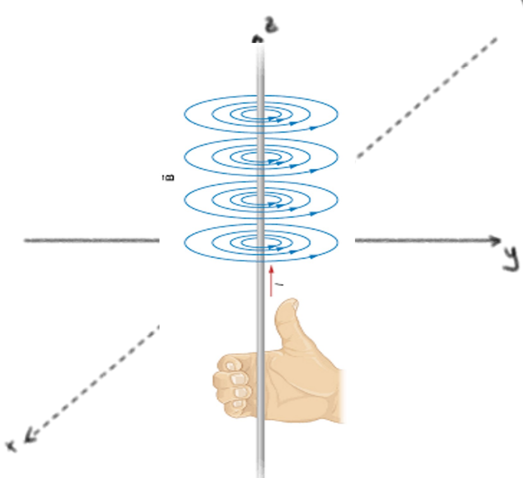


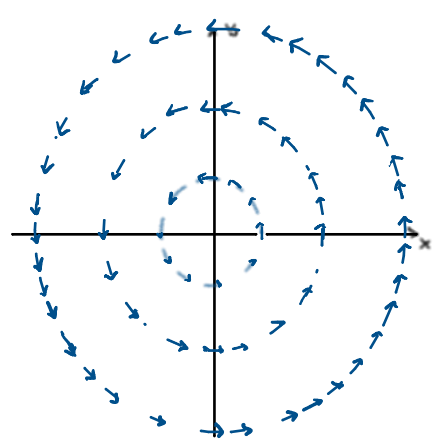
(2) Electric field due to point charge

$$\vec{E} = \frac{kQ}{(x^2+y^2+z^2)^{3/2}} (x\hat{\imath} + y\hat{\jmath} + z\hat{k})$$

(3) magnetic field due to an infinite current carrying wire along the z-axis:

$$\vec{B}(x,y,z) = \frac{AJ_z}{\sqrt{x^2+y^2}}(-y\hat{\imath} + x\hat{\jmath} + 0\hat{k})$$





9.3 We are often interested in how much a field "pierces" through a volume (divergence) and how much rotation it locally induces (curl).

10_ Divergence

10.1

To begin, let $\mathbf{v}(x, y, z)$ be a differentiable vector function, where x, y, z are Cartesian coordinates, and let v_1, v_2, v_3 be the components of $\mathbf{v}$. Then the function

(1) $$\operatorname{div} \mathbf{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

is called the **divergence** of $\mathbf{v}$ or the *divergence of the vector field defined by* $\mathbf{v}$.

10.2 Example.

Find the divergence of $\vec{v}$ at $P(1, 0, -1)$ for

$$\vec{v} = \frac{1}{(x^2+y^2+z^2)^{3/2}} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Soln: Start with:

$$\frac{\partial}{\partial x} v_1 = \frac{\partial}{\partial x}\left(\frac{x}{(x^2+y^2+z^2)^{3/2}}\right)$$

$$= \frac{(x^2+y^2+z^2)^{3/2}\frac{\partial}{\partial x}x - x\frac{\partial}{\partial x}(x^2+y^2+z^2)^{3/2}}{\left[(x^2+y^2+z^2)^{3/2}\right]^2}$$

$$= \frac{(x^2+y^2+z^2)^{3/2} - x\left[\frac{3}{2}(x^2+y^2+z^2)^{1/2}(2x)\right]}{(x^2+y^2+z^2)^3}$$

$$= \frac{(x^2+y^2+z^2)^{1/2}\left[(x^2+y^2+z^2) - 3x^2\right]}{(x^2+y^2+z^2)^3}$$

$$= \frac{y^2+z^2-2x^2}{(x^2+y^2+z^2)^{3-1/2}} = \frac{y^2+z^2-2x^2}{(x^2+y^2+z^2)^{5/2}}$$

Slido: What will you get for $\frac{\partial}{\partial y}v_2$ and $\frac{\partial}{\partial z}v_3$?

what do you get for $\frac{\partial}{\partial y} v_2$ and $\frac{\partial}{\partial z} v_3$?

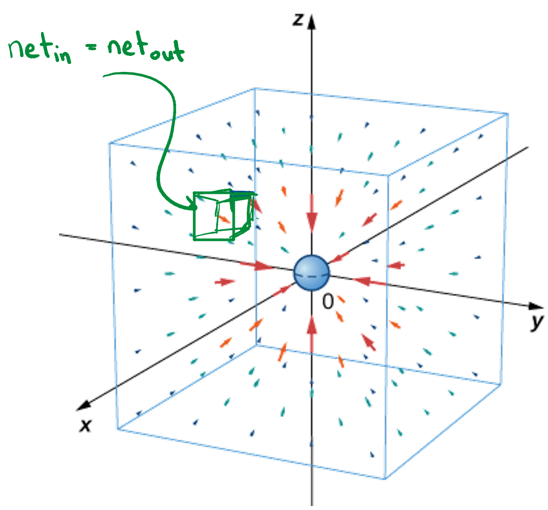
Then divergence of $\vec{v}$ is:

$$\vec{\nabla}\cdot\vec{v} = \frac{\partial}{\partial x}v_1 + \frac{\partial}{\partial y}v_2 + \frac{\partial}{\partial z}v_3$$

$$= \frac{y^2+z^2-2x^2}{(x^2+y^2+z^2)^{5/2}} + \frac{x^2+z^2-2y^2}{(x^2+y^2+z^2)^{5/2}} + \frac{y^2+x^2-2z^2}{(x^2+y^2+z^2)^{5/2}}$$

$$= 0 \qquad \text{(no divergence!)}$$

10.3 Meaning of $\vec{\nabla}\cdot\vec{v} = 0$?



Related to continuity equation:

$$\frac{\partial}{\partial t}\rho + \vec{\nabla}\cdot(\rho\vec{v}) = 0$$

$\vec{\nabla}\cdot\vec{v} = 0$ is known as the condition of incompressibility

Roughly speaking, the divergence measures outflow minus inflow.

10.3 Example Find $\vec{\nabla}\cdot\vec{v}$ at $P(2, \pi/2, 0)$ for

$$\vec{v} = [0,\ \cos(xyz),\ \sin(xyz)]$$

Soln: We have:

$$\frac{\partial}{\partial x}0 = 0$$

$$\frac{\partial}{\partial y}\cos(xyz) = -\sin(xyz)(xz) = -xz\sin(xyz)$$

$$\frac{\partial}{\partial z}\sin(xyz) = \cos(xyz)(xy) = xy\cos(xyz)$$

Thus:

$$\vec{\nabla}\cdot\vec{v} = 0 - xz\sin(xyz) + xy\cos(xyz)$$

$$= x\left[y\cos(xyz) - z\sin(xyz)\right]$$

Then $\vec{\nabla}\cdot\vec{v}\big|$ ... $2[\ \ (2\ \pi\ 0)\ \ (2\ \pi\ 0)]$

$$= x\left[y\cos(xyz) - z\sin(xyz)\right]$$

Then $\vec{\nabla}\cdot\vec{v}\Big|_{(2,\pi/2,0)} = 2\left[\frac{\pi}{2}\cos\left(2\cdot\frac{\pi}{2}\cdot 0\right) - 0\sin\left(2\cdot\frac{\pi}{2}\cdot 0\right)\right]$

$$= 2\left(\frac{\pi}{2}\cos(0) - 0\right) = 2\left(\frac{\pi}{2}\right) = \pi$$

10.4 Important Properties of Divergence:

(a) $\operatorname{div}(k\mathbf{v}) = k \operatorname{div}\mathbf{v}$ (k constant)

(b) $\operatorname{div}(f\mathbf{v}) = f\operatorname{div}\mathbf{v} + \mathbf{v}\bullet\nabla f$

(c) $\operatorname{div}(f\nabla g) = f\nabla^2 g + \nabla f\bullet\nabla g$

(d) $\operatorname{div}(f\nabla g) - \operatorname{div}(g\nabla f) = f\nabla^2 g - g\nabla^2 f$

10.5

THEOREM 1

Invariance of the Divergence

The divergence div $\mathbf{v}$ *is a scalar function, that is, its values depend only on the points in space (and, of course, on* $\mathbf{v}$*) but not on the choice of the coordinates in* (1), *so that with respect to other Cartesian coordinates* x^*, y^*, z^* *and corresponding components* v_1^*, v_2^*, v_3^* *of* $\mathbf{v}$,

(2) $$\operatorname{div}\mathbf{v} = \frac{\partial v_1^*}{\partial x^*} + \frac{\partial v_2^*}{\partial y^*} + \frac{\partial v_3^*}{\partial z^*}.$$

10.6 What is the theorem saying? Slido:

If we change coordinates, then the formula for quantities such as $\vec{\nabla}f$ and $\vec{\nabla}\cdot\vec{v}$ change. However, $\vec{\nabla}\cdot\vec{v}$ at some point P is just a number, and it doesn't change regardless of the coordinates used.

10.7 Example: We know Cartesian $\longleftrightarrow$ Polar is defined by $x = r\cos\theta$, $y = r\sin\theta$

So the point P with Cartesian Coordinates $(1,1)$ has polar coordinates $(\sqrt{2}, \pi/4)$.

Suppose $\vec{v} = (x^2+y^2)\hat{\imath} - y/x\,\hat{\jmath}$. so that in Cartesian:

$$\vec{\nabla}\cdot\vec{v} = \frac{\partial}{\partial x}(x^2+y^2) - \frac{\partial}{\partial y}\left(\frac{y}{x}\right) = (2x) - 1/x = \frac{2x^2-1}{x}$$

at $x=1$, $y=1$: $\vec{\nabla}\cdot\vec{v}\Big|_P = \frac{2(1)^2-1}{1} = 1$

In Polar Coordinates:

$$\vec{v} = r^2\hat{\imath} - \tan\theta\,\hat{\jmath} = r^2\left(\cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}\right) - \tan\theta\left(\sin\theta\,\hat{r} + \cos\theta\,\hat{\theta}\right)$$

$$\Rightarrow \vec{v} = \left[r^2\cos\theta - \sin\theta\tan\theta\right]\hat{r} - \left[r^2\sin\theta + \sin\theta\right]\hat{\theta}$$

$$\vec{v} = r^2\hat{i} - \tan\theta\,\hat{j} = r^2\left(\cos\theta\,\hat{r} - \sin\theta\,\hat{\theta}\right) - \tan\theta\left(\ldots\right)$$

$$\Rightarrow \vec{v} = \left[r^2\cos\theta - \sin\theta\tan\theta\right]\hat{r} - \left[r^2\sin\theta + \sin\theta\right]\hat{\theta}$$

Then divergence in Polar Coordinates is computed as:

$$\vec{\nabla}\cdot\vec{v} = \frac{1}{r}\frac{\partial}{\partial r}\left[r\left(r^2\cos\theta - \sin\theta\tan\theta\right)\right] + \frac{1}{r}\frac{\partial}{\partial\theta}(-\sin\theta)(r^2+1)$$

$$= \frac{1}{r}\left[3r^2\cos\theta - \sin\theta\tan\theta\right] - \frac{(r^2+1)}{r}\cos\theta$$

at $r=\sqrt{2}$, $\theta=\pi/4$

$$\vec{\nabla}\cdot\vec{v} = \frac{1}{\sqrt{2}}\left[3(\sqrt{2})^2\underbrace{\cos(\pi/4)}_{1/\sqrt{2}} - \underbrace{\sin(\pi/4)}_{1/\sqrt{2}}\underbrace{\tan(\pi/4)}_{1}\right] - \frac{((\sqrt{2})^2+1)}{\sqrt{2}}\underbrace{\cos(\pi/4)}_{1/\sqrt{2}}$$

$$\vec{\nabla}\cdot\vec{v} = \frac{1}{\sqrt{2}}\left[3(2)\left(\frac{1}{\sqrt{2}}\right) - \frac{1}{\sqrt{2}}\right] - \frac{(3)}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}$$

$$= \frac{1}{\sqrt{2}}\left(\frac{5}{\sqrt{2}}\right) - \frac{3}{2} = \frac{5}{2} - \frac{3}{2}$$

$$= \frac{2}{2} = 1$$

As expected, we get the same number, even though the calculation is hair-raisingly challenging.

11 – The Laplacian

11.1 For a scalar field f, the divergence of its gradient, that is

$$\vec{\nabla}\cdot(\vec{\nabla}f) \text{ is written as}$$

$$\nabla^2 f = \frac{\partial^2}{\partial x^2}f + \frac{\partial^2}{\partial y^2}f + \frac{\partial^2}{\partial z^2}f$$

and is known as the Laplacian of f.

11.2 Example: Calculate $\nabla^2 f$ for $f = e^{xyz}$

Soln: $\frac{\partial}{\partial x}f = \frac{\partial}{\partial x}e^{xyz} = yz\,e^{xyz}$

Thus $\frac{\partial}{\partial x}\left(\frac{\partial}{\partial x}f\right) = \frac{\partial}{\partial x}\left(yz\,e^{xyz}\right) = (yz)^2 e^{xyz} \leftarrow \frac{\partial^2}{\partial x^2}f$

Similarly: $\frac{\partial^2}{\partial y^2}f = (xz)^2 e^{xyz}$ and $\frac{\partial^2}{\partial z^2}f = (xy)^2 e^{xyz}$

Then

$$\nabla^2 f = (yz)^2 e^{xyz} + (xz)^2 e^{xyz} + (xy)^2 e^{xyz}$$

Then

$$\nabla^2 f = (yz)^2 e^{xyz} + (xz)^2 e^{xyz} + (xy)^2 e^{xyz}$$

$$= e^{xyz} \left[y^2 z^2 + x^2 z^2 + x^2 y^2 \right]$$